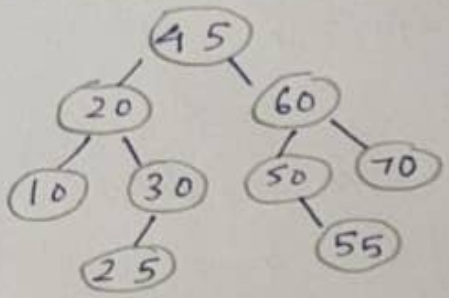
2) An online bookstore stores the following ISBN keys: 45, 20, 60, 10, 30, 50, 70, 25, 55.

(a) Construct a BST by inserting the ISBNs in the given order. Execute the program and display the resulting tree using inorder, preorder and postorder traversals.

ans.



Inorder - 10, 20, 25, 30, 45, 50, 55, 60, 70

Preorder - 45, 20, 10, 30, 25, 60, 50, 55, 70

Postorder - 10, 25, 30, 20, 55, 50, 70, 60, 45

(b) Search for 25, 55 and 90 using:
BST search
Linear search
Record the number of comparisons required for each search.

ans. (1) BST search

Search for 25:

40 → 20 → 30 → 25

Comparisons = 4.

Search for 55:

45 → 60 → 50 → 55

Comparisons = 4.

Search for 90:

45 → 60 → 70 → NULL

Comparisons = 3

90 is not present.

(b) Linear Search

Search 25:

45 → 20 → 60 → 10 → 30 → 50 → 70 → 25

Comparisons = 8

Search 55:

45 → 20 → 60 → 10 → 30 → 50 → 70 → 25 → 55

Comparisons = 9.

Search 90:

45 → 20 → 60 → 10 → 30 → 50 → 70 → 25 → 55 → not found

Comparisons = 9.

(c) Analyse how the shape and height of the BST influence search performance. Compare the observed number of comparisons with the theoretical complexity of BST search and linear search. Conclude which method could be preferable for the given dataset and why.

ans. Shape and Height of BST :-

The BST has 9 nodes and its height is 3 edges (4 levels). Since the tree is reasonably balanced, searching requires fewer comparisons than linear search for this dataset.

Theoretical complexity

Operation	BST	Linear Search
Search - Average	$O(\log n)$	$O(n)$
Search - Worst case	$O(n)$	$O(n)$
Space	$O(n)$	$O(1)$

For a balanced BST, the search path is short. However if a BST becomes highly skewed, its height can become n-1, making its search performance similar to linear search.

Conclusion.

For the given dataset, BST search requires fewer comparisons than linear search for 25, 55 and 90. Therefore, a BST is preferable when frequent searching is required, especially when the tree remains reasonably balanced.

Observation:

BST → 4, 4, 3 comparisons.

linear search → 8, 9, 9 comparisons.